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Investigating the Role of PrP^{c} in PrP^{Sc} Seeding Using CRISPR-Cas9 and RT-QuIC: A Quantitative Analysis of PrP^{c}-Dependent Seeding

Introduction

Prion diseases, or transmissible spongiform encephalopathies (TSEs), represent a class of universally fatal neurodegenerative disorders in humans and animals, including Creutzfeldt-Jakob disease (CJD) in humans, scrapie in sheep and goats, and bovine spongiform encephalopathy (BSE) in cattle [1], [2] Biologically,prions occupy a unique space in molecular pathology: unlike conventional infectious agents such as viruses or bacteria, prions propagate through protein-mediated conformational template replication without requiring nucleic acids [1], [2].The central mechanism driving prion neurotoxicity is the post-translational misfolding of the host-encoded cellular prion protein (PrP^{c}) into an abnormal, insoluble,β-sheet-rich isoform termed pathogenic scrapie prion (PrP^{Sc}) [1], [2].Exogenous or newly formed PrP^{Sc} acts as a conformational seed, binding to the endogenous native PrP^{c} and catalyzing its structural conversion into self-propagating amyloid aggregates [2].

Because endogenous PrP^{c} serves as the homologous substrate for disease propagation and host toxicity, its presence is susceptible to prion infection [2], [3]. Landmark studies in genetic knockout models demonstrated that *Prnp*-devoid($Prnp^{0/0}$) mice are entirely resistant to scrapie infection and fail to propagate or transmit PrP^{Sc} [3].Furthermore, heterozygous $Prnp^{+/0}$ animals exhibiting partial PrP^{c} depletion show significantly prolonged disease incubation periods,supporting a direct dose-dependent relationship between endogenous PrP^{c} expression and disease progression [3]. Subsequent therapeutic strategies including transient siRNA-mediated knockdowns [4], [5], peptide-mediated silencing [5],modern live-animal knockout models [6], and antisense oligonucleotides have further demonstrated that reducing PrP^{c} can delay disease progression and limit prion propagation.

However, historical RNA interference (RNAi) and siRNA approaches suffer from key experimental limitations,including incomplete protein clearance,transient knockdown kinetics, and off-target sequence effects [4], [5]. Precision gene-editing technologies,specifically CRISPR-Cas9, provide a tool to achieve permanent,targeted, and gene-dose-specific disruptions of the human/murine PRNP locus [6].By generating stable cell lines with precise, quantified levels of residual PrP^{C}, researchers can dissect sub-stoichiometric dynamics of prion misfolding that were previously inaccessible using incomplete or transient knockdown techniques [4], [5], [6].

Evaluating sub-stoichiometric PrP^{c} concentrations requires high-sensitivity platforms like Real-Time Quaking-Induced Conversion (RT-QuIC), an ultrasensitive Seed amplification Assay that uses cyclic shaking and Thioflavin T fluorescence to detect minute PrP^{Sc} seeds [7], [8], [9].Beyond binary diagnostics [7], [9]. RT-QuIC kinetics including lag time aggregation velocity, and peak fluorescence allow quantitative modelling of conversion thresholds and energy barriers as substrate levels drop [8].

Research question and hypothesis

This pilot study aims to answer these research questions:

1. What reduction in endogenous PrP^c expression can be achieved through CRISPR-Cas9 disruption of PRNP in the selected cell line?

This quantitative research question investigates the extent to which CRISPR-Cas9 mediated disruption of the PRNP gene is associated with reduced endogenous PrP^c expression. The independent variable is the CRISPR-Cas9 mediated PRNP disruption condition, while the dependent variable is the level of residual endogenous cellular PrP^c expression [5], [6].

2.How does the degree of PrP^cdepletion affect PrP^{Sc}seeding susceptibility in cell lysates?

This quantitative research question investigates whether the degree of PrP^c depletion is associated with changes in PrP^{Sc} seeding susceptibility. Previous studies have shown that reduced PrP^c levels can delay prion disease progression and that PrP^c availability influences prion replication,supporting its role as a limiting substrate for propagation [3],[4]. The independent variable is the degree of residual PrP^c expression,while the dependent variable [1] is PrP^{Sc} seeding susceptibility.

3.What relationship exists between residual PrP^clevels and RT-QuIC amplification kinetics?

This quantitative research question investigates whether residual PrP^clevels are associated with measurable differences in RT-QuIC amplification kinetics. Previous research has established RT-QuIC as a sensitive method for detecting prion seeding activity, while kinetic measurements can provide quantitative information about amplification behavior [7]-[9]. The independent variable is the level of residual PrP^c, while the dependent variable is RT-QuIC amplification kinetics.

We hypothesize that CRISPR-Cas9-mediated disruption of PRNP will reduce residual PrP^c levels, resulting in decreased PrP^{Sc} seeding susceptibility and delayed RT-QuIC amplification kinetics [3]- [9].

Literature Review

Prion diseases are fatal neurodegenerative disorders characterized by the template-directed misfolding of cellular prion protein (PrP^c) into a pathogenic isoform (PrP^{Sc}). Investigating the therapeutic depletion of host PrP^c alongside the analytical performance of Real-Time Quaking-Induced Conversion (RT-QuIC) provides a framework for evaluating both substrate reduction and diagnostic sensitivity.

1. Therapeutic Substrate Depletion (PrP^C)

Because PrP^c serves as the obligatory substrate for propagation, suppressing its expression represents a primary therapeutic strategy.

- **Gene Knockout:** Büeler *et al.* [3] demonstrated that PrP^c-knockout mice (Prn-$p^{0/0}$) are completely resistant to scrapie infection without developing severe developmental defects.

- **RNA Interference:** H.Bender *et al.* [4] achieved transient PrP^c knockdown *in vivo* using liposome-siRNA-peptide complexes (LSPCs), extending survival and preserving cognitive function in infected mice.
- **CRISPR-Cas9 Ablation:** Translating substrate reduction to gene editing , Mehrabian *et al*. [5] applied CRISPR-Cas9 to ablate PRNP *in vitro*, confirming knockout without disrupting the broader cellular proteome.
- **In Vivo Translation:** Allais-Bonnet *et al*. [6] expanded this approach to livestock models, demonstrating successful *in vivo* CRISPR/Cas9 editing of PRNP in Alpine goats.

2. Analytical Efficacy and Technical Constraints of RT-QuIC

To monitor seeding activity following PrP^c depletion, cell-free amplification assays must reliably detect sub-picogram amounts of PrP^{Sc}.

- **Diagnostic Performance & Kinetic Analysis:** Koníčková *et al.* [8] highlighted that RT-QuIC offers exceptional analytical sensitivity and specificity across neurodegenerative conditions, noting that kinetic parameters (lag time,slope,max fluorescence) provide quantitative seed data beyond binary readouts. Shi *et al.* [7] further established quantitative RT-QuIC (qRT-QuIC) for precise titration of tissue seed concentrations.
- **Protocol Standardizations & Assay Robustness:** Orrù *et al.* [9] validated RT-QuIC as a robust diagnostic platform across human prion diseases, demonstrating reproducible amplification when using standardized recombinant substrate (rPrP).

3. Conflicting Evidence: Is RT-QuIC Fully Effective?

Despite its broad adoption, literature reveals critical operational limitations that impact whether RT-QuIC functions as an absolute quantitative tool:

1. **Spontaneous Aggregation vs. True Seeding:** Cheng *et al.* [10] demonstrated that physical and chemical factors such as shaking speed, temperature, and buffer composition heavily influenced rPrP stability. Under certain optimized conditions designed to increase speed, rPrP can undergo false-positive spontaneous aggregation in the absence of PrP^{Sc} seeds. This creates a trade-off between assay speed and diagnostic specificity.
2. **Matrix Effects and Inhibitors:** While Orrù *et al.* [9] demonstrated high fidelity in clean biological matrices (e.g.,CSF), raw tissue homogenates or heavily edited cellular lysates contain endogenous lipid and protein inhibitors that prolong the lag phase [10], complicating kinetic interpretations in knockdown models.
3. **Phenotypic Trade-offs of Substrate Depletion:** While initial knockout models [3] suggested PrP^c depletion is completely benign, subsequent studies (e.g., Bremer *et al*.,2010) found that long-term ablation induces demyelinating peripheral neuropathy,demonstrating that complete targeted deletion carries biological side effects [11].

4. Research Gap

While literature establishes that total PrP^{c} ablation halts prion propagation [1], [3] and that RT-QuIC reliably amplifies minute PrP^{Sc} seeds [7], [9], a critical gap remains: no study has systematically quantified how incremental, partial CRISPR-mediated PrP^{C} depletion directly alters RT-QuIC amplification kinetics. Current research treats gene editing primarily as a binary knockout [3], [5] and RT-QuIC as a diagnostic threshold assay [8], [9], leaving the exact mathematical relationship between residual host substrate levels and specific kinetic parameters $(\Delta t, k, F_{max})$ unresolved [8], [10].

Research Methodology

This study uses a quantitative experimental design to evaluate PRNP knockout efficiency, PrP^{Sc} seeding susceptibility, and RT-QuIC amplification kinetics in a cell culture model.

1. ***PRNP* Gene Editing and Expression Profiling (RQ1)**
   Mouse neuroblastoma (N2a) cells are transfected with Cas9 ribonucleoproteins targeting *PRNP* exon 3 [5]. Clones are isolated into discrete experimental groups: **Control**(100% expression), **Partial Knockdown**(targeted 25%,50%, and 75% depletion), and **Complete Knockout**(0% expression) [3], [5]. Residual PrP^{c} levels are quantified via sandwich ELISA and normalized using Western blot analysis against β-actin loading controls [4], [5].
2. **Cell Lysate Preparation & Seeding (RQ2)**
   Cell lysates (1 mg/mL) from each experimental group are prepared using non-denaturing lysis buffer and spiked with defined RML scrapie brain homogenate seeds [3], [4]. Protease-resistant PrP^{Sc} formation is verified following Proteinase K (PK) digestion and Western blot analysis to determine how varying tiers of host PrP^{c} reduction affect initial seeding susceptibility [3], [4], [11].
3. **RT-QuIC Assays & Kinetic Tracking (RQ3)**
   Multi-well reaction plates containing recombinant protein substrate ($rPrP^{90-231}$), Thioflavin T (ThT) fluorescent dye, and serial dilutions (10^{-2} to 10^{-8}) of the spiked cell lysates are incubated at 42°C with automated 60-second shaking and 60-second rest cycles [9], [10]. Key kinetic parameters tracked via real-time fluorescence reading include [8], [10]:

- **Lag time(Δt):** The total hours required for ThT fluorescence to cross the detection threshold [8], [10].
- **Amplification Rate (k):** The slope of the exponential log-phase reaction curve [7], [8].
- **Maximum Fluorescence (F_{max}):** The plateau capacity representing total amyloid fibril yield [7], [8], [10].

4. Statistical Analysis & Hypothesis Testing

- **ANOVA & Post-Hoc Analysis:** One-way Analysis of Variance (ANOVA) with Tukey's post-hoc test compares mean PrP^C reduction across CRISPR treatment groups(RQ1) [5] and evaluates significant differences in lag time and F_{max} across knockdown tiers(RQ2,RQ3) [4], [8].
- **Non-Linear Regression:** Four-parameter logistic (4PL) sigmoidal curve fitting models ThT fluorescence over time [7], [8]. Spearman's rank correlation coefficient (r_s) determines the statistical strength between percent PrP^c depletion and lag time extension [7].
- **Chi-Square (χ^2) Tests**: A Chi-square (χ^2)goodness-of-fit test compares observed positive/negative conversion frequencies across serial dilutions against expected theoretical distribution models [7], [8]. A Chi-square test of independence evaluates whether non-specific spontaneous substrate aggregation occurs independently of seed presence [10].
- **Significance:** All tests will use a significance threshold of α=0.05 processed in GraphPad Prism or R.

Timeline:

- *Weeks 1-2:* Transfect N2a cells with Cas9 RNPs targeting *PRNP* exon 3 [5].
- *Weeks 3-4:* Isolate single-cell clones (Control 100%,25%,50%,75%,0% Knockout) and quantify residual PrP^C via ELISA and normalized Western blot [3],[4],[5].
- *Week 5:* Harvest lysates (1 mg/mL) across groups and spike with defined RML scrapie seeds [3],[4].
- *Week 6:* Perform Proteinase K (PK) digestion and Western blot analysis to verify initial seeding susceptibility [3],[4],[11].
- *Week 7:* Set up 96-well plates with $rPrP^{90-231}$ substrate, ThT dye, and lysate serial dilutions (10^{-2} to 10^{-8}) [9],[10].
- *Week 8:* Run assays at 42°C with automated shaking/rest cycles, recording fluorescence every 45 min to measure Lag Time,Amplification Rate, and Max Fluorescence [7],[8],[10].
- *Weeks 9-10:* Fit 4PL sigmoidal curves, run one-way ANOVA with Tukey's post-hoc tests, and perform Chi-square goodness-of-fit and independence tests on kinetics and spontaneous aggregation rates [4],[5],[7],[8],[10].
- *Weeks 11-12:* Revise the content, check citations and finalize everything before submitting.

Conclusion: By quantifying how partial *PRNP* depletion alters RT-QuIC kinetics(Δt,k,Fmax), this study establishes a mathematical framework for substrate-reduction therapeutics [5], [8]. Identifying optimal knockdown thresholds will clarify if partial protein reduction can halt prion propagation, bypassing the phenotypic risks of complete *PRNP* knockout [1], [3], [7].

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